File

Name: R4-D1-D2-270-stepped-hub
Description: KISSsoft example
Changed by: AAA on: 31.10.2026 at: 16:51:11

1 Overview

Calculation method: DIN 6892-B:2012

Label		DIN 6885.1:1968 Default
Key width (mm)	[b]	32.00
Key height (mm)	[h]	18.00
Chamfer (mean value) (mm)	[r]	0.70
Shaft diameter (mm)	[d]	120.00
Nominal torque (Nm)	[T]	4000.00
Application factor	[KA]	1.50
equivalent torque (Nm)	[Teq]	6000.00
Maximum torque (Nm)	[Tmax]	15000.00
Minimal frictional torque for interference fit (Nm)	[TRmin]	0.00
Torque curve: With alternating torque		
Maximum backwards torque (Nm)	[TmaxR]	15000.00
Number of load peaks	[NL]	10000
Number of change of load direction	[NW]	250000
Load direction changing coefficient	[fw]	0.58
Number of keys	[i]	1
Load factor	[phi]	1.00
equivalent circumferential stress (N)	[Feq]	100000.00
Maximal circumferential force (N)	[Fmax]	250000.00
Contact coefficient for equivalent surface pressure	[Kveq]	1.00
Contact coefficient for maximal surface pressure	[Kvmax]	1.00
Help coefficient	[Kλe]	1.085
Load distribution coefficient	[Kλ]	1.085
Friction factor for equivalent torsional moment	[KReq]	1.000
Friction factor for maximum peak torsional moment	[KRmax]	1.000

2 Shaft

Material, own Input		1 C 60 N
Type		Through hardened steel
Treatment		unalloyed/through hardened
Tensile strength (component's raw diameter) (N/mm²)	[Rm]	600.00
Yield point (component's raw diameter) (N/mm²)	[Rp]	310.00
Component raw diameter (mm)	[deff]	130
Groove depth shaft (minimal value) (mm)	[t1]	11.00
Chamfer on shaft (mm)	[s1]	0.00
Supporting key length (mm)	[ltr]	93.00
Load-bearing shaft groove depth (mm)	[t1tr]	8.13
Equivalent effective surface pressure (N/mm²)	[peq]	143.55
Maximum effective surface pressure (N/mm²)	[pmax]	358.87
Support factor	[fs]	1.30
Hardness influence coefficient	[fH]	1.00

Permissible surface pressure (N/mm²)	[pzul]	403.00
Load peak frequency coefficient	[fL]	1.40
Load direction changing coefficient	[fw]	0.58
Safety factor, equivalent load (fw pzul/peq)	[SFeq]	1.62
Safety, maximum load (fL pzul/pmax)	[SFmax]	1.57
Minimal safety	[SF]	1.57

3 Hub

Material, own Input	EN GJL-250	
Type	GJL (Cast iron flake graphite)	
Treatment	untreated	
Tensile strength (wall thickness) (N/mm²)	[Rm]	200.00
Yield point (wall thickness) (N/mm²)	[Rp]	130.00
The hub material strength values are defined using the wall thickness as the raw diameter.		
Component raw diameter (wall thickness) (mm)	[deff]	80
Groove depth hub (minimal value) (mm)	[t2]	7.40
Chamfer on hub (mm)	[s2]	0.80
Supporting key length (mm)	[ltr]	93.00
Load-bearing hub groove depth (mm)	[t2tr]	7.90
Small outside diameter of hub (mm)	[D1]	200.00
Big outside diameter of hub (mm)	[D2]	270.00
Width of hub-part with D2 (mm)	[c]	17.00
Equivalent diameter hub (mm)	[D]	206.95
Distance a0 (Figure 2, DIN 6892) (mm)	[a0]	96.00
Equivalent effective surface pressure (N/mm²)	[peq]	147.68
Maximum effective surface pressure (N/mm²)	[pmax]	369.20
Support factor	[fs]	2.00
Hardness influence coefficient	[fH]	0.00
Permissible surface pressure (N/mm²)	[pzul]	400.00
Load peak frequency coefficient	[fL]	1.20
Load direction changing coefficient	[fw]	0.58
Safety factor, equivalent load (fw pzul/peq)	[SFeq]	1.56
Safety, maximum load (fL pzul/pmax)	[SFmax]	1.30
Minimal safety	[SF]	1.30

4 Key

Material, own Input	C45 K			
Type	Through hardened steel			
Treatment	unalloyed/through hardened			
Tensile strength (N/mm²)	[Rm]	680.00		
Yield point (N/mm²)	[Rp]	430.00		
Equivalent effective surface pressure (N/mm²)	[peq]	143.55	/	147.68
Maximum effective surface pressure (N/mm²)	[pmax]	358.87	/	369.20
Support factor	[fs]	1.10		
Hardness influence coefficient	[fH]	1.00		
Permissible surface pressure (N/mm²)	[pzul]	473.00		
Load peak frequency coefficient	[fL]	1.40		
Load direction changing coefficient	[fw]	0.58		
Safety factor, equivalent load (fw pzul/peq)	[SFeq]	1.85		
Safety, maximum load (fL pzul/pmax)	[SFmax]	1.79		
Minimal safety	[SF]	1.79		
Cross section area (mm²)	[b*ltr]	2976.00		
Shear stress (N/mm²)	[tau]	33.60		

5 Remarks

Safety = minimum (fw pzul/peq, fL pzul/pmax)
Condition according to DIN 6892 Safety >= 1.0
Chamfer on key: Mean value as in examples in DIN 6892
Groove depth: Minimum value as in examples in DIN 6892

End of report (lines: 115)
